Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “average” to explain your model or result. Sentence starters are optional.

impulse: net force acting over time; change in momentum

average: one value representing an interval or a set of values

momentum: mass times velocity, including direction

01 Warm-up: bring back an earlier idea

A planet has surface field 75 N/kg. At center distance 5 R, find field and the force on a 7 kg object.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

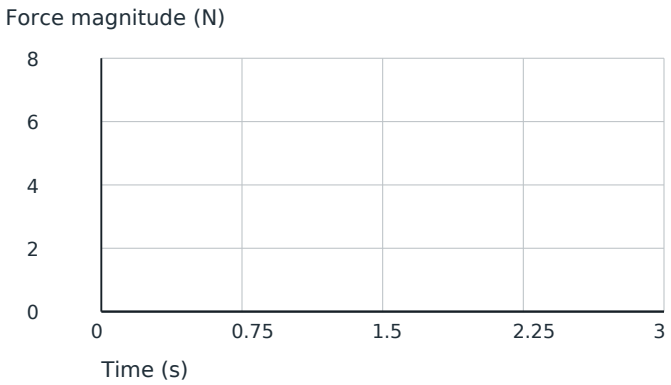
02 Worked example: explain the model

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant 6 N force acts right for 2 s. Impulse is 12 N·s right, so momentum increases by 12 kg·m/s.

Your turn: Draw the force-time rectangle and shade its area.
Explain why the two impulse units are equivalent.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

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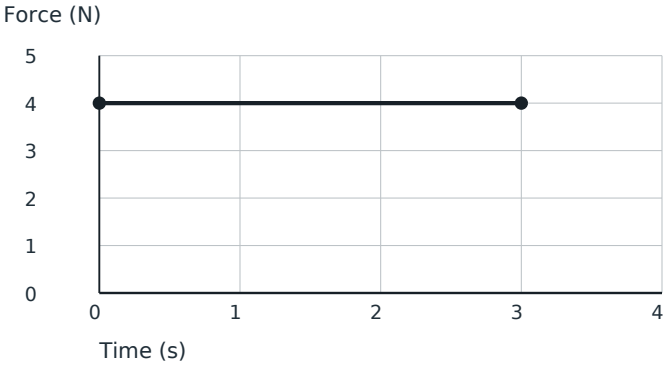
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Read the rectangular force-time graph.
Find the area and state what it means.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart starts at rest and receives 6 N·s rightward impulse. Final momentum is 6 kg·m/s.
Find final velocity by dividing by mass.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	**After**

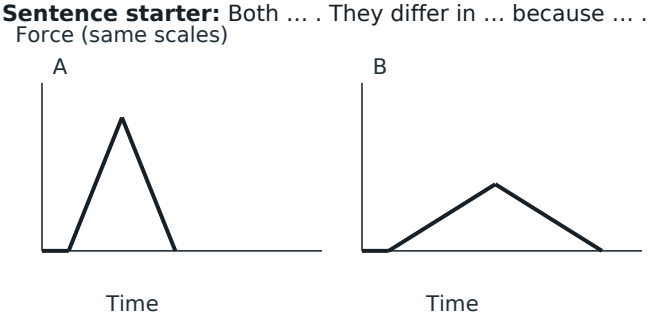
Label mass and velocity.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Sentence starter: Both ... . They differ in ... because ... .



Same scales: compare impulse and peak force.

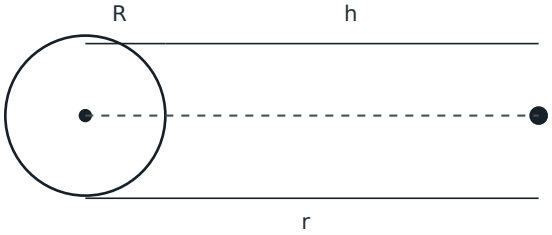
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A spherical planet has surface field 144 N/kg.
Find the field at center distance 6 R, where R is the planet radius.
Label the masses, field or force relevant to the question.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: My design shows ... . It meets the requirement by ... .



Marzano check - circle: ① ② ③

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Continue down each column.

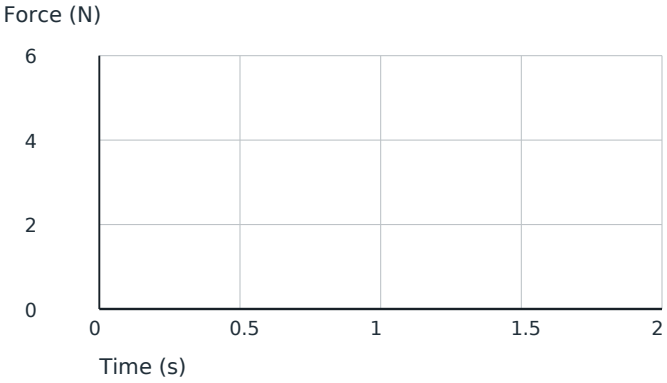
Sentence starters are optional. You may plan in any language.

07 Exit check

A 5 N force acts right for 2 s.
Find impulse. On a 2 kg cart initially at rest, find final velocity.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

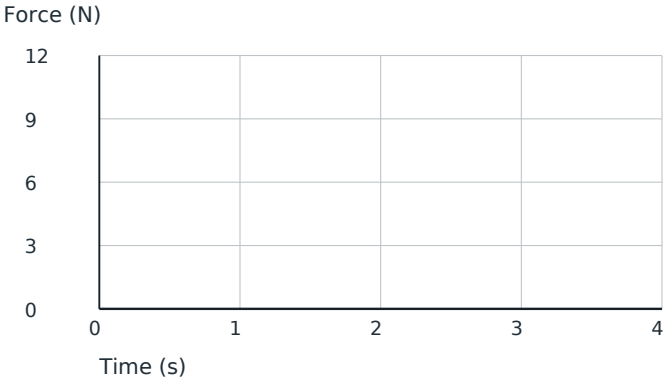


Marzano check - circle: 1 2 3

08 STRETCH: test the model

Create two constant force-and-time pairs giving an impulse of 12 N·s.
Compare their force-time rectangles.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3